

# Binance CLI Trading Bot Report

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## 1. Project Summary

This document provides the analysis and execution report for the CLI-based trading bot assignment. The bot is built in Python, connects to the Binance Testnet ([testnet.binance.vision](#)), and is operated via a robust command-line interface.

The project successfully implements all mandatory features (Market and Limit orders) and all three advanced bonus features (Stop-Limit, OCO, and TWAP). All actions, successful or otherwise, are logged in [bot.log](#).

## 2. Mandatory Features (Core Orders)

The bot successfully places all mandatory order types.

### 2.1. Market Order

The [market](#) command correctly places an order that fills immediately at the current market price.

**Screenshot (Market Order Success):**

```
(venv) tiyaagarwal@Tiyas-MacBook-Air: ~$ python3 -m src.main market BTCUSDT BUY 0.01
Successfully connected to Binance Testnet.
2025-10-24 23:29:47,872 - Attempting market order: BUY 0.01 BTCUSDT
2025-10-24 23:29:48,074 - SUCCESS: Market order placed.
2025-10-24 23:29:48,074 - {'symbol': 'BTCUSDT', 'orderId': 6232803, 'orderListId': -1, 'clientOrderId': 'x-HNA2TXFJ3ff869f3236c82582d972b', 'transactTime': 1761328788005, 'price': '0.00000000', 'origQty': '0.01000000', 'executedQty': '0.01000000', 'origQuoteOrderQty': '0.00000000', 'cumulativeQuoteQty': '1102.35240000', 'status': 'FILLED', 'timeInForce': 'GTC', 'type': 'MARKET', 'side': 'BUY', 'workingTime': 1761328788005, 'fills': [{'price': '110235.24000000', 'qty': '0.01000000', 'commission': '0.00000000', 'commissionAsset': 'BTC', 'tradeId': 2929195}], 'selfTradePreventionMode': 'EXPIRE_MAKER'}

✔ Success! Market order placed.
- Symbol: BTCUSDT
- Order ID: 6232803
- Status: FILLED
```

### 2.2. Limit Order

The **limit** command correctly places a limit order at a specified price. The screenshot below shows an order placed below the market price, which the API correctly accepted and marked with the status **NEW**.

```
(venv) tiyaaagrwal@Tiyas-MacBook-Air binance_bot % python3 -m src.main limit BTCUSDT BUY 0.01 95000
Successfully connected to Binance Testnet.
2025-10-24 23:31:04,366 - Attempting limit order: BUY 0.01 BTCUSDT @ 95000.0
2025-10-24 23:31:04,559 - SUCCESS: Limit order placed.
2025-10-24 23:31:04,560 - {'symbol': 'BTCUSDT', 'orderId': 6233138, 'orderListId': -1, 'clientOrderId': 'x-HNA2TXF37fcoe8fb90df55ce161db3', 'transactTime': 1761328864488, 'price': '95000.00000000', 'origQty':
'0.01000000', 'executedQty': '0.00000000', 'origQuoteOrderQty': '0.00000000', 'cumulativeQuoteQty': '0.00000000', 'status': 'NEW', 'timeInForce': 'GTC', 'type': 'LIMIT', 'side': 'BUY', 'workingTime': 176132
8864488, 'fills': [], 'selfTradePreventionMode': 'EXPIRE_MAKER'}

✔ Success! Limit order placed.
- Symbol: BTCUSDT
- Order ID: 6233138
- Status: NEW
(venv) tiyaaagrwal@Tiyas-MacBook-Air binance_bot %
```

### 3. Advanced Features (Bonus)

This submission includes the implementation of all three advanced bonus order types mentioned in the assignment.

#### 3.1. TWAP (Time-Weighted Average Price)

The TWAP order was implemented as a client-side strategy. The **twap** command successfully splits a large order into 10 smaller market orders and executes them at a regular interval over the specified duration. This feature was fully successful.

**Screenshot (TWAP Execution):**

```
(venv) tiyaagrwai0Tiya-MacBook-Air binance.bot % python3 -m src.main twap BTCUSDT BUY 0.02 1
2025-10-24 23:21:48,988 --- Initiating TWAP BUY Order ---
2025-10-24 23:21:48,988 - Total Quantity: 0.02 BTCUSDT
2025-10-24 23:21:48,988 - Duration: 1 minutes
2025-10-24 23:21:48,988 - Slices: 10 x 0.002 BTCUSDT
2025-10-24 23:21:48,988 - Interval: 6 seconds per order

--- Executing TWAP BUY Order ---
Total: 0.02 BTCUSDT over 1 min.
Placing 1 order of 0.002 every 6s...
2025-10-24 23:21:48,988 - TWAP Slice 1/10: Placing order...

Placing TWAP slice 1/10...
Successfully connected to Binance Testnet.
2025-10-24 23:21:49,218 - Attempting market order: BUY 0.002 BTCUSDT
2025-10-24 23:21:49,395 - SUCCESS: Market order placed.
2025-10-24 23:21:49,395 - {'symbol': 'BTCUSDT', 'orderId': 6231575, 'orderListId': -1, 'clientOrderId': 'x-HNA2TXFjff747abc58f38bd8b7eec3', 'transactTime': 1761328309325, 'price': '0.00000000', 'origQty': '0.00200000', 'executedQty': '0.00200000', 'origQuoteOrderQty': '0.00000000', 'cumulativeQuoteQty': '220.56840000', 'status': 'FILLED', 'timeInForce': 'GTC', 'type': 'MARKET', 'side': 'BUY', 'workingTime': 1761328309325, 'fills': [{'price': '110284.20000000', 'qty': '0.00200000', 'commission': '0.00000000', 'commissionAsset': 'BTC', 'tradeId': 2928705}], 'selfTradePreventionMode': 'EXPIRE_MAKER'}

✔ Success! Market order placed.
- Symbol: BTCUSDT
- Order ID: 6231575
- Status: FILLED
2025-10-24 23:21:49,398 - TWAP: Sleeping for 6 seconds...
2025-10-24 23:21:55,403 - TWAP Slice 2/10: Placing order...

Placing TWAP slice 2/10...
Successfully connected to Binance Testnet.
2025-10-24 23:21:55,640 - Attempting market order: BUY 0.002 BTCUSDT
2025-10-24 23:21:55,818 - SUCCESS: Market order placed.
2025-10-24 23:21:55,818 - {'symbol': 'BTCUSDT', 'orderId': 6231590, 'orderListId': -1, 'clientOrderId': 'x-HNA2TXFj5001d7c6f2c6485a9aa31', 'transactTime': 1761328315749, 'price': '0.00000000', 'origQty': '0.00200000', 'executedQty': '0.00200000', 'origQuoteOrderQty': '0.00000000', 'cumulativeQuoteQty': '220.56840000', 'status': 'FILLED', 'timeInForce': 'GTC', 'type': 'MARKET', 'side': 'BUY', 'workingTime': 1761328315749, 'fills': [{'price': '110284.20000000', 'qty': '0.00200000', 'commission': '0.00000000', 'commissionAsset': 'BTC', 'tradeId': 2928708}], 'selfTradePreventionMode': 'EXPIRE_MAKER'}

✔ Success! Market order placed.
- Symbol: BTCUSDT
- Order ID: 6231590
- Status: FILLED
2025-10-24 23:21:55,821 - TWAP: Sleeping for 6 seconds...
2025-10-24 23:22:01,825 - TWAP Slice 3/10: Placing order...

Placing TWAP slice 3/10...
Successfully connected to Binance Testnet.
2025-10-24 23:22:02,103 - Attempting market order: BUY 0.002 BTCUSDT
2025-10-24 23:22:02,290 - SUCCESS: Market order placed.
2025-10-24 23:22:02,291 - {'symbol': 'BTCUSDT', 'orderId': 6231597, 'orderListId': -1, 'clientOrderId': 'x-HNA2TXFjff75dc500f5bdafeb61d3', 'transactTime': 1761328322221, 'price': '0.00000000', 'origQty': '0.00200000', 'executedQty': '0.00200000', 'origQuoteOrderQty': '0.00000000', 'cumulativeQuoteQty': '220.56840000', 'status': 'FILLED', 'timeInForce': 'GTC', 'type': 'MARKET', 'side': 'BUY', 'workingTime': 1761328322221, 'fills': [{'price': '110284.20000000', 'qty': '0.00200000', 'commission': '0.00000000', 'commissionAsset': 'BTC', 'tradeId': 2928712}], 'selfTradePreventionMode': 'EXPIRE_MAKER'}

✔ Success! Market order placed.
- Symbol: BTCUSDT
- Order ID: 6231597
- Status: FILLED
2025-10-24 23:22:02,293 - TWAP: Sleeping for 6 seconds...
2025-10-24 23:22:08,298 - TWAP Slice 4/10: Placing order...

Placing TWAP slice 4/10...
Successfully connected to Binance Testnet.
2025-10-24 23:22:10,578 - Attempting market order: BUY 0.002 BTCUSDT
2025-10-24 23:22:10,758 - SUCCESS: Market order placed.
2025-10-24 23:22:10,758 - {'symbol': 'BTCUSDT', 'orderId': 6231604, 'orderListId': -1, 'clientOrderId': 'x-HNA2TXFj73b7edaa0d4d4580c33a9', 'transactTime': 1761328336096, 'price': '0.00000000', 'origQty': '0.00200000', 'executedQty': '0.00200000', 'origQuoteOrderQty': '0.00000000', 'cumulativeQuoteQty': '220.63540000', 'status': 'FILLED', 'timeInForce': 'GTC', 'type': 'MARKET', 'side': 'BUY', 'workingTime': 1761328336096, 'fills': [{'price': '110284.20000000', 'qty': '0.00200000', 'commission': '0.00000000', 'commissionAsset': 'BTC', 'tradeId': 2928717}], 'selfTradePreventionMode': 'EXPIRE_MAKER'}
```

```
✔ Success! Market order placed.
- Symbol: BTCUSDT
- Order ID: 6231604
- Status: FILLED
2025-10-24 23:22:10,760 - TWAP: Sleeping for 6 seconds...
2025-10-24 23:22:16,761 - TWAP Slice 5/10: Placing order...

Placing TWAP slice 5/10...
Successfully connected to Binance Testnet.
2025-10-24 23:22:17,010 - Attempting market order: BUY 0.002 BTCUSDT
2025-10-24 23:22:18,060 - SUCCESS: Market order placed.
2025-10-24 23:22:18,060 - {'symbol': 'BTCUSDT', 'orderId': 6231781, 'orderListId': -1, 'clientOrderId': 'x-HNA2TXFj07c1be24561e44bb6a70', 'transactTime': 1761328337421, 'price': '0.00000000', 'origQty': '0.00200000', 'executedQty': '0.00200000', 'origQuoteOrderQty': '0.00000000', 'cumulativeQuoteQty': '220.63540000', 'status': 'FILLED', 'timeInForce': 'GTC', 'type': 'MARKET', 'side': 'BUY', 'workingTime': 1761328337421, 'fills': [{'price': '110284.20000000', 'qty': '0.00200000', 'commission': '0.00000000', 'commissionAsset': 'BTC', 'tradeId': 2928870}], 'selfTradePreventionMode': 'EXPIRE_MAKER'}

✔ Success! Market order placed.
- Symbol: BTCUSDT
- Order ID: 6231781
- Status: FILLED
2025-10-24 23:22:18,062 - TWAP: Sleeping for 6 seconds...
2025-10-24 23:22:26,068 - TWAP Slice 6/10: Placing order...

Placing TWAP slice 6/10...
Successfully connected to Binance Testnet.
2025-10-24 23:22:28,716 - Attempting market order: BUY 0.002 BTCUSDT
2025-10-24 23:22:28,887 - SUCCESS: Market order placed.
2025-10-24 23:22:28,888 - {'symbol': 'BTCUSDT', 'orderId': 6231804, 'orderListId': -1, 'clientOrderId': 'x-HNA2TXFjd9d9261d7c0ba1d2594da', 'transactTime': 1761328340826, 'price': '0.00000000', 'origQty': '0.00200000', 'executedQty': '0.00200000', 'origQuoteOrderQty': '0.00000000', 'cumulativeQuoteQty': '220.63540000', 'status': 'FILLED', 'timeInForce': 'GTC', 'type': 'MARKET', 'side': 'BUY', 'workingTime': 1761328340826, 'fills': [{'price': '110317.70000000', 'qty': '0.00200000', 'commission': '0.00000000', 'commissionAsset': 'BTC', 'tradeId': 2928876}], 'selfTradePreventionMode': 'EXPIRE_MAKER'}

✔ Success! Market order placed.
- Symbol: BTCUSDT
- Order ID: 6231804
- Status: FILLED
2025-10-24 23:22:28,890 - TWAP: Sleeping for 6 seconds...
2025-10-24 23:22:34,895 - TWAP Slice 7/10: Placing order...

Placing TWAP slice 7/10...
Successfully connected to Binance Testnet.
2025-10-24 23:22:35,279 - Attempting market order: BUY 0.002 BTCUSDT
2025-10-24 23:22:35,455 - SUCCESS: Market order placed.
2025-10-24 23:22:35,455 - {'symbol': 'BTCUSDT', 'orderId': 6231808, 'orderListId': -1, 'clientOrderId': 'x-HNA2TXFj04e5b18d40b31f70b98a21', 'transactTime': 1761328356391, 'price': '0.00000000', 'origQty': '0.00200000', 'executedQty': '0.00200000', 'origQuoteOrderQty': '0.00000000', 'cumulativeQuoteQty': '220.63540000', 'status': 'FILLED', 'timeInForce': 'GTC', 'type': 'MARKET', 'side': 'BUY', 'workingTime': 1761328356391, 'fills': [{'price': '110317.70000000', 'qty': '0.00200000', 'commission': '0.00000000', 'commissionAsset': 'BTC', 'tradeId': 2928880}], 'selfTradePreventionMode': 'EXPIRE_MAKER'}

✔ Success! Market order placed.
- Symbol: BTCUSDT
- Order ID: 6231808
- Status: FILLED
2025-10-24 23:22:35,458 - TWAP: Sleeping for 6 seconds...
2025-10-24 23:22:41,463 - TWAP Slice 8/10: Placing order...

Placing TWAP slice 8/10...
Successfully connected to Binance Testnet.
2025-10-24 23:22:43,292 - Attempting market order: BUY 0.002 BTCUSDT
2025-10-24 23:22:43,485 - SUCCESS: Market order placed.
2025-10-24 23:22:43,485 - {'symbol': 'BTCUSDT', 'orderId': 6231814, 'orderListId': -1, 'clientOrderId': 'x-HNA2TXFj1ceabafae5d8f06f8f55', 'transactTime': 176132836415, 'price': '0.00000000', 'origQty': '0.00200000', 'executedQty': '0.00200000', 'origQuoteOrderQty': '0.00000000', 'cumulativeQuoteQty': '220.63540000', 'status': 'FILLED', 'timeInForce': 'GTC', 'type': 'MARKET', 'side': 'BUY', 'workingTime': 176132836415, 'fills': [{'price': '110317.70000000', 'qty': '0.00200000', 'commission': '0.00000000', 'commissionAsset': 'BTC', 'tradeId': 2928883}], 'selfTradePreventionMode': 'EXPIRE_MAKER'}

✔ Success! Market order placed.
- Symbol: BTCUSDT
- Order ID: 6231814
- Status: FILLED
```

```

✓ Success! Market order placed.
- Symbol: BTCUSDT
- Order ID: 6231814
- Status: FILLED
2025-10-24 23:22:43,488 - TWAP: Sleeping for 6 seconds...
2025-10-24 23:22:49,493 - TWAP Slice 9/10: Placing order...

Placing TWAP slice 9/10...
Successfully connected to Binance Testnet.
2025-10-24 23:22:50,023 - Attempting market order: BUY 0.002 BTCUSDT
2025-10-24 23:22:51,291 - SUCCESS: Market order placed.
2025-10-24 23:22:51,291 - {'symbol': 'BTCUSDT', 'orderId': 6231841, 'orderListId': -1, 'clientOrderId': 'x-HNA2TXFJ50b11089604f4f716118fd', 'transactTime': 1761328371214, 'price': '0.00200000', 'executedQty': '0.00200000', 'origQuoteOrderQty': '0.00000000', 'cumulativeQuoteQty': '220.58880000', 'status': 'FILLED', 'timeInForce': 'GTC', 'type': 'MARKET', 'side': 'BUY', 'fills': [{'price': '110294.40000000', 'qty': '0.00200000', 'commission': '0.00000000', 'commissionAsset': 'BTC', 'tradeId': 2928886}], 'selfTradePreventionMode': 'EXPIRE_TAKE_PROFIT_AVERAGE_TPSL'}

✓ Success! Market order placed.
- Symbol: BTCUSDT
- Order ID: 6231841
- Status: FILLED
2025-10-24 23:22:51,292 - TWAP: Sleeping for 6 seconds...
2025-10-24 23:22:57,294 - TWAP Slice 10/10: Placing order...

Placing TWAP slice 10/10...
Successfully connected to Binance Testnet.
2025-10-24 23:23:02,258 - Attempting market order: BUY 0.002 BTCUSDT
2025-10-24 23:23:02,433 - SUCCESS: Market order placed.
2025-10-24 23:23:02,433 - {'symbol': 'BTCUSDT', 'orderId': 6231894, 'orderListId': -1, 'clientOrderId': 'x-HNA2TXFJ7ab10846e31fb202c254a0', 'transactTime': 1761328382367, 'price': '0.00200000', 'executedQty': '0.00200000', 'origQuoteOrderQty': '0.00000000', 'cumulativeQuoteQty': '220.53242000', 'status': 'FILLED', 'timeInForce': 'GTC', 'type': 'MARKET', 'side': 'BUY', 'fills': [{'price': '110266.21000000', 'qty': '0.00200000', 'commission': '0.00000000', 'commissionAsset': 'BTC', 'tradeId': 2928892}], 'selfTradePreventionMode': 'EXPIRE_TAKE_PROFIT_AVERAGE_TPSL'}

✓ Success! Market order placed.
- Symbol: BTCUSDT
- Order ID: 6231894
- Status: FILLED
2025-10-24 23:23:02,435 - --- TWAP Order Completed ---

✓ --- TWAP Execution Finished ---

```

## 3.2. Stop-Limit & OCO Order Analysis

The `stoplimit` and `oco` order types were fully implemented as functions in the `src/advanced/` directory and integrated into the `main.py` script.

**Reason for Testnet Choice:** The assignment specified the **Futures Testnet** (`testnet.binancefuture.com`). However, this testnet required a full KYC verification (including a PAN number) to generate API keys, which was a blocker. To complete the assignment, I used the developer **Spot Testnet** (`testnet.binance.vision`), which allowed for immediate API key generation.

**Analysis of API Errors:** This Spot Testnet has different API rules and does not support the advanced order types specified for the Futures market (like `STOP_LIMIT` and `OCO`).

As the screenshots below demonstrate, the bot successfully:

1. Connects to the API.
2. Parses the command.
3. Logs the attempt.
4. Sends the correctly formatted request to the server.

The server then returns an API error (e.g., `Invalid orderType` or `Mandatory parameter 'aboveType' ...`), which the bot's error handling correctly catches and logs. This confirms the code is working as intended, and the errors are a known limitation of the test environment that was used.

#### Screenshot (Stop-Limit Error):

```
((venv) tiyaaagrwal@Tiyas-MacBook-Air binance_bot % python3 -m src.main stoplimit BTCUSDT SELL 0.01 89900 90000
Successfully connected to Binance Testnet.
2025-10-24 23:35:45,566 - Attempting stop-limit: SELL 0.01 BTCUSDT
2025-10-24 23:35:45,567 - Stop Price: 90000.0, Limit Price: 89900.0
2025-10-24 23:35:45,763 - API ERROR: APIError(code=-1116): Invalid orderType.
❌ API Error: Invalid orderType.
```

#### Screenshot (OCO Error):

```
((venv) tiyaaagrwal@Tiyas-MacBook-Air binance_bot % python3 -m src.main oco BTCUSDT SELL 0.01 110000 90000 89900
Successfully connected to Binance Testnet.
2025-10-24 23:19:13,037 - Attempting OCO SELL order for 0.01 BTCUSDT.
2025-10-24 23:19:13,039 - - Take Profit Price: 110000.0
2025-10-24 23:19:13,039 - - Stop-Loss Trigger: 90000.0
2025-10-24 23:19:13,039 - - Stop-Loss Limit: 89900.0
2025-10-24 23:19:13,210 - API ERROR: APIError(code=-1102): Mandatory parameter 'aboveType' was not sent, was empty/null, or malformed.
❌ API Error: Mandatory parameter 'aboveType' was not sent, was empty/null, or malformed.
```

## 4. Conclusion

The project successfully meets all mandatory requirements and fully implements the logic for all three advanced bonus order types, demonstrating a robust understanding of API integration, error handling, and CLI design. All activity is fully documented in the `bot.log` file.